

# Optimizing Hydrogen Systems and Demand Response for Enhanced Integration of RES and EVs in Smart Grids

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**Abstract**—Global interest in maximizing the hosting capacity (HC) of renewable energy sources (RES) while avoiding high energy curtailment has lately grown in smart grids (SGs). This trend has been associated with the empowerment of hydrogen systems (HS) and demand response (DR). In this regard, this article proposes a new approach to optimally synergize HS and DR for elevated integration of intermittent RES in SGs. The proposed approach is predicated on a coordinated management scheme for various SG control devices, including: a complete HS (hydrogen tank, water electrolyzer, and fuel cell), hydrogen load DR, RES energy curtailment, DSTATCOM functionality of the RES inverters, storage charging rates, and electrical DR. Additionally, the flexibility of electric vehicles (EVs) with G2V/V2G functionalities has been considered to support RES and HS expansion. The proposed approach is formulated as a multi-objective optimization model that aims to optimize two competing objective functions, i.e., total costs and HC capacity of RESs. The simulation results reveal the superiority of the proposed approach which resulted in a notable HC increase of up to 17.24%, accompanied by a significant cost reduction. The proposed approach is also applied to define the optimal number of HS to maximize RES capacity.

**Index Terms**—Demand response, hosting capacity, hydrogen system, renewable energy, smart grid.

## NOMENCLATURE

### Abbreviations

|     |                                            |
|-----|--------------------------------------------|
| CRF | capital recovery factor for various assets |
| DR  | demand response                            |
| FC  | fuel cell                                  |
| HC  | hosting capacity                           |

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|     |                          |
|-----|--------------------------|
| HDR | hydrogen demand response |
| HS  | hydrogen system          |
| HT  | hydrogen tank            |
| LF  | levelization factor      |
| PV  | photovoltaic             |
| SoC | state of charge          |
| WE  | water electrolyzer       |
| WT  | wind turbine             |

### List of Sets

|              |                      |
|--------------|----------------------|
| $\mathbb{B}$ | set of buses         |
| $\mathbb{L}$ | set of states        |
| $\mathbb{N}$ | set of EVs           |
| $\mathbb{T}$ | set of time segments |

### List of Parameters

|                                   |                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $\alpha, \beta$                   | Beta pdf parameters                                                                                                           |
| $\eta^{\text{FC}}$                | efficiency of the FC                                                                                                          |
| $\eta^{\text{WE}}$                | efficiency of the WE                                                                                                          |
| $\mathcal{D}$                     | discount rate (%)                                                                                                             |
| $\mathcal{E}$                     | escalation factor (%)                                                                                                         |
| $\mathcal{P}^{\mathcal{R}}(t, y)$ | probability of the solar irradiance at time $t$ and state $y$                                                                 |
| $\mathcal{P}^d(t, l)$             | probability of the load demand at time $t$ and state $l$                                                                      |
| $\mathcal{P}^v(t, w)$             | probability of the wind speed at time $t$ and state $w$                                                                       |
| $\mathcal{X}$                     | matrix of three columns that include all possible combinations of the PV, WT, and load demand states at each time instant $t$ |
| $\mathfrak{B}$                    | complete annual generation-load model                                                                                         |
| $\mathfrak{R}$                    | interest rate of the asset (%)                                                                                                |
| $\mu_{\mathcal{R}}(t)$            | mean of the solar irradiance at time $t$                                                                                      |
| $\mu_d(t)$                        | mean of the load demand at time $t$                                                                                           |
| $\mu_v(t)$                        | mean of the wind speed at time $t$                                                                                            |
| $\Pi(t, \ell)$                    | combined probability matrix of RES and load demand                                                                            |
| $\pi^g(t, \ell)$                  | price of the grid electricity (\$/kWh)                                                                                        |
| $\pi^{\text{ch}}$                 | cost of curtailed hydrogen power (\$/kWh)                                                                                     |
| $\pi^{\text{curt}}$               | cost of curtailed RES power (\$/kWh)                                                                                          |
| $\pi^{\text{dr}}$                 | discomfort cost of DR (\$/kWh)                                                                                                |
| $\pi^{\text{EV}}$                 | O&M cost of EVs chargers (\$/kWh)                                                                                             |
| $\pi^{\text{h2p}}$                | O&M cost of FC units (\$/kWh)                                                                                                 |

|                                          |                                                                 |                                   |                                                                                             |
|------------------------------------------|-----------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------|
| $\pi^{\text{hdr}}$                       | discomfort cost of HDR (\$/kWh)                                 | $\mu^m$                           | normalized membership function of $m^{\text{th}}$ solution                                  |
| $\pi^{\text{p2h}}$                       | O&M cost of WE units (\$/kWh)                                   | $\mu_\zeta$                       | membership function of the objective function $\zeta$                                       |
| $\pi^{\text{PV}}$                        | O&M cost of PV units (\$/kWh)                                   | $C^{\text{inv}}$                  | total annual investment cost (\$/year)                                                      |
| $\pi^{\text{WT}}$                        | O&M cost of WT units (\$/kWh)                                   | $C^{\text{ope}}(t, \ell)$         | total operation cost for each $t$ and $\ell$                                                |
| $\pi^h$                                  | hydrogen price (\$/kWh)                                         | $D_i^{\text{H,DR}}(t, \ell)$      | hydrogen demand at bus $i$ for each $t$ and $\ell$ after applying HDR (kW)                  |
| $\text{Nd}(t, \ell)$                     | number of days included in a typical time $t$ and state $\ell$  | $f_\zeta^{\text{max}}$            | maximum value of function of the objective function $\zeta$                                 |
| $\sigma_{\mathcal{R}}(t)$                | standard deviation of the solar irradiance at time $t$          | $f_\zeta^{\text{min}}$            | minimum value of function of the objective function $\zeta$                                 |
| $\sigma_d(t)$                            | standard deviation of the load demand at time $t$               | $f_\zeta$                         | value of function of the objective function $\zeta$                                         |
| $\sigma_v(t)$                            | standard deviation of the wind speed at time $t$                | $I_i^{\text{H2P}}(t, \ell)$       | binary variables for HS discharging                                                         |
| $c, k$                                   | scale and shape parameters of Weibull PDF                       | $I_i^{\text{P2H}}(t, \ell)$       | binary variables for HS charging                                                            |
| $C^{\text{FC}}$                          | capacity of a FC unit (kW)                                      | $I_{i,k}^{\text{ch,EV}}(t, \ell)$ | binary variables for $k^{\text{th}}$ EV charging at bus $i$                                 |
| $C^{\text{HT}}$                          | capacity of a HT unit (kW)                                      | $I_{i,k}^{\text{dc,EV}}(t, \ell)$ | binary variables for $k^{\text{th}}$ EV discharging at bus $i$                              |
| $C^{\text{inv,HS}}$                      | investment costs of HS (\$/kWh)                                 | $N_i^{\text{FC}}$                 | number of FC units installed at bus $i$                                                     |
| $C^{\text{inv,PV}}$                      | investment costs of PV (\$/kW)                                  | $N_i^{\text{HT}}$                 | number of HT units installed at bus $i$                                                     |
| $C^{\text{inv,WT}}$                      | investment costs of WT (\$/kW)                                  | $N_i^{\text{PV}}$                 | number of PV units installed at bus $i$                                                     |
| $C^{\text{PV}}$                          | capacity of a PV unit (kW)                                      | $N_i^{\text{WE}}$                 | number of WE units installed at bus $i$                                                     |
| $C^{\text{WE}}$                          | capacity of a WE unit (kW)                                      | $N_i^{\text{WT}}$                 | number of WT units installed at bus $i$                                                     |
| $C^{\text{WT}}$                          | capacity of a WT unit (kW)                                      | $P^g(t, \ell)$                    | grid power for each $t$ and $\ell$ (kW)                                                     |
| $D_i^H(t, \ell)$                         | hydrogen demand at bus $i$ for time $t$ and state $\ell$ (kW)   | $P_i^H(t, \ell)$                  | Amount of purchased hydrogen from hydrogen network at bus $i$ for each $t$ and $\ell$ (kW)  |
| $f^b(t, \mathcal{R})$                    | beta pdf of solar irradiance $\mathcal{R}$ at time instant $t$  | $P_i^{\text{CH}}(t, \ell)$        | curtailed hydrogen at bus $i$ for each $t$ and $\ell$ (kW)                                  |
| $f^n(t, d)$                              | normal pdf of load demand $d$ at time instant $t$               | $P_i^{\text{DR}}(t, \ell)$        | DR power at bus $i$ for each $t$ and $\ell$ (kW)                                            |
| $f^r(t, v)$                              | Rayleigh pdf of wind speed $v$ at time instant $t$              | $P_i^{\text{H2P}}(t, \ell)$       | FC generated power at bus $i$ for each $t$ and $\ell$ (kW)                                  |
| $N^{\text{ope}}$                         | operation period (years)                                        | $P_i^{\text{HDR}}(t, \ell)$       | HDR at bus $i$ for each $t$ and $\ell$ (kW)                                                 |
| $N^{\text{yr}}$                          | loan term (years)                                               | $P_i^{\text{P2H}}(t, \ell)$       | WE power at bus $i$ for each $t$ and $\ell$ (kW)                                            |
| $N_{\text{nd}}$                          | number of non-dominated solutions                               | $P_i^{\text{Pvc}}(t, \ell)$       | curtailed PV power at bus $i$ for each $t$ and $\ell$ (kW)                                  |
| $N_{\text{obj}}$                         | number of objective functions                                   | $P_i^{\text{PV}}(t, \ell)$        | PV generated power at bus $i$ for each $t$ and $\ell$ (kW)                                  |
| $P_i^d(t, \ell)$                         | electrical demand at bus $i$ for each $t$ and $\ell$ (kW)       | $P_i^{\text{Wc}}(t, \ell)$        | curtailed WT power at bus $i$ for each $t$ and $\ell$ (kW)                                  |
| $P_{\text{max}}^{\text{DR}}$             | maximum DR factor                                               | $P_i^{\text{WT}}(t, \ell)$        | WT generated power at bus $i$ for each $t$ and $\ell$ (kW)                                  |
| $P_{\text{max}}^{\text{HDR}}$            | maximum HDR factor                                              | $P_i^{d,\text{DR}}(t, \ell)$      | electrical demand at bus $i$ for each $t$ and $\ell$ after applying DR (kW)                 |
| $P_{i,\text{max}}^{\text{H2P}}(t, \ell)$ | maximum FC power at bus $i$ for each $t$ and $\ell$ (kW)        | $P_{i,k}^{\text{ch,EV}}(t, \ell)$ | charging power of $k^{\text{th}}$ EV at $i^{\text{th}}$ for each $t$ and $\ell$ (kW)        |
| $P_{i,\text{max}}^{\text{P2H}}(t, \ell)$ | maximum WE power at bus $i$ for each $t$ and $\ell$ (kW)        | $P_{i,k}^{\text{dc,EV}}(t, \ell)$ | discharging power of $k^{\text{th}}$ EV at $i^{\text{th}}$ bus for each $t$ and $\ell$ (kW) |
| $P_{i,\text{min}}^{\text{H2P}}(t, \ell)$ | minimum FC power at bus $i$ for each $t$ and $\ell$ (kW)        | $Q_i^g(t, \ell)$                  | generated reactive power at bus $i$ for each $t$ and $\ell$ (kvar)                          |
| $P_{i,\text{min}}^{\text{P2H}}(t, \ell)$ | minimum WE power at bus $i$ for each $t$ and $\ell$ (kW)        | $Q_i^{\text{PV}}(t, \ell)$        | reactive power of PV at bus $i$ for each $t$ and $\ell$ (kvar)                              |
| $Q_i^d(t, \ell)$                         | reactive power demand at bus $i$ for each $t$ and $\ell$ (kvar) | $Q_i^{\text{WT}}(t, \ell)$        | reactive power of WT at bus $i$ for each $t$ and $\ell$ (kvar)                              |
| $\text{HSoc}_i^{\text{initial}}$         | initial SOC of the hydrogen tank at bus $i$                     | $V_i(t, \ell)$                    | voltage at bus $i$ for each $t$ and $\ell$ (pu)                                             |
| $\text{Pb}_i^{\text{FC}}$                | principal borrowed of FC installed at bus $i$ (\$/kW)           | $\text{HSoc}_i(t, \ell)$          | SoC of the hydrogen tank at bus $i$ for each $t$ and $\ell$                                 |
| $\text{Pb}_i^{\text{HT}}$                | principal borrowed of HT installed at bus $i$ (\$/kW)           | $\text{SoC}_{i,k}(t, \ell)$       | SoC of $k^{\text{th}}$ EV at bus $i$ for each $t$ and $\ell$                                |
| $\text{Pb}_i^{\text{PV}}$                | principal borrowed of PV installed at bus $i$ (\$/kW)           |                                   |                                                                                             |
| $\text{Pb}_i^{\text{WE}}$                | principal borrowed of WE installed at bus $i$ (\$/kW)           |                                   |                                                                                             |
| $\text{Pb}_i^{\text{WT}}$                | principal borrowed of WT installed at bus $i$ (\$/kW)           |                                   |                                                                                             |
| $\mathcal{G}_1$                          | the annual cost objective function                              |                                   |                                                                                             |
| $\mathcal{G}_2$                          | the HC objective function                                       |                                   |                                                                                             |

## I. INTRODUCTION

**R**ECENTLY, more and more renewable energy sources (RESs), mainly wind and photovoltaic generation systems, are promoted to be integrated into smart grids (SGs). In this context, hydrogen systems (HS) and demand response can play important roles in developing the capabilities of SGs. Specifically, HS, as a clean and versatile energy source and storage, have a potential feature to address intermittency challenges associated with RES generation [1], [2], [3]. The utilization of surplus electricity to produce hydrogen through water electrolysis technology has the capability to increase renewable energy penetration towards sustainable SGs. By leveraging advanced technologies to seamlessly integrate hydrogen production, storage, and utilization, a dynamic solution for balancing supply and demand can be optimally achieved in SGs. Concurrently, demand response mechanisms can empower consumers to actively participate in grid optimization by adjusting their energy consumption. Therefore, the synergy between HS and demand response not only enhances grid reliability but also contributes to international decarbonization efforts [4], [5], [6]. Most importantly, the hosting capacity (HC) of RES can be maximized without curtailing excessive energy in SGs.

In the literature, several research studies have been directed to develop planning models for RES considering HS from diverse aspects. In [7], the authors have incorporated the electric network into the interregional hydrogen supply chains due to its ability to economically provide energy at long distances. In [8], a planning algorithm for an electricity-hydrogen unified energy system has been introduced considering hydrogen production and storage abilities. [9], a collaborative planning model for both electricity and transportation systems is introduced to achieve the overall emission reduction in both systems considering hydrogen fuel cell vehicles. The authors of [10] have utilized particle swarm optimization for the optimal planning of a combined nuclear-renewable micro-hybrid energy system considering hydrogen storage. Reference [11] has introduced an optimal operation and design approach that considers system reliability with various renewable energy levels while incorporating emerging devices e.g. electrolysis cells and hydrogen storage tanks. In [12], power-to-gas devices and plug-in hybrid electric and hydrogen-powered vehicles have been utilized to form a hydrogen-electricity energy-sharing model. In [13], an optimal operation of hydrogen refueling stations has been proposed to provide frequency ancillary services to the power grid. The authors of [14] have addressed the coordinated planning of high-voltage direct current and power-to-hydrogen supply chains for renewable energy utilization. The authors of [15] have proposed a coordination planning model of an integrated energy system and electric vehicles for reducing carbon emissions. In [16], an optimal planning model is proposed for the dual-zero microgrid where the seasonal shift of energy flow is accomplished by the hydrogen storage system. In [17], an improved particle swarm optimization has been applied to solve a planning model that includes different energy generation players, including hydrogen storage systems, photovoltaic generation, and wind turbines.

Other research studies have focused on empowering the role of DR programs for RES planning in modern energy systems. In [18], an optimal electricity-heat system planning model has been proposed to enhance the accommodation of wind energy while considering an integrated DR model. The authors of [19] have introduced a blockchain-based DR management that provides optimal energy trading between electric vehicles (EVs) and charging stations. In [20], a day-ahead energy management model has been proposed for microgrids with a DR aggregator as an intermediate coordinator to optimize the social welfare of microgrids while considering the privacy of end-users. The authors of [21] have formulated a stochastic-risk model for microgrid participation in various services markets considering DR while modeling uncertainties of solar radiation and wind speed. In [22], a novel mixed-integer linear programming model has been formulated for integrated generation, storage, and DR expansion planning while capturing flexibility dynamics. A risk-aware Volt/VAR support model integrated with a reinforcement learning control scheme has been proposed in [23] for unbalanced distribution systems while considering DR and voltage reduction strategies. In [24], a planning model has been proposed to identify the ideal DR intensity for multi-carrier microgrids, which can save energy and provide financial profits via tuning energy demands during peak instant. The authors of [25] have proposed a coordinative planning model of public transport electrification and RESs where public transport hubs are modeled as flexible nodes with DR features. The authors of [26] have integrated DR with a holistic energy management system with PV generation and energy storage considering commercial customers.

The integration of HS, DR, and EVs to maximize the HC of RES faces several key challenges [27], [28], [29]. Coordinating diverse components like hydrogen tanks (HT), water electrolyzers (WE), fuel cells (FC), and EVs with grid interaction features demands advanced control and communication systems [27]. Scaling such systems for larger grids with diverse demands presents further challenges. High upfront costs, regulatory complexities, and maintaining grid stability amidst renewable energy variability add to the difficulty [30]. Managing data, gaining public acceptance, and building technical expertise also pose significant challenges [31]. Therefore, to effectively address these challenges, several key approaches can be utilized. Adopting a holistic energy system planning approach that considers different assets enables the efficient utilization and advancement of RES within the smart grid [32]. Also, conducting comprehensive techno-economic analysis across the entire hydrogen supply chain allows for informed decision-making by evaluating economic, environmental, and societal factors, thus maximizing overall benefits while minimizing risks [33]. Moreover, implementing energy management systems tailored to industrial clusters minimizes operational costs and maximizes green hydrogen production, enhancing the economic viability of sustainable energy solutions [34]. These strategies collectively contribute to the transition towards a greener and more efficient energy landscape while addressing key challenges in integrating complex systems into diverse grids.

TABLE I  
MAJOR FEATURES OF PROPOSED AND PREVIOUSLY  
PLANNING APPROACHES

| REF  | RES |    | Unce. | DSTATCOM | DR  |     | EVs | APC |
|------|-----|----|-------|----------|-----|-----|-----|-----|
|      | PV  | WT |       |          | LDR | HDR |     |     |
| [7]  | ✓   | ✓  | -     | -        | -   | ✓   | -   | -   |
| [8]  | ✓   | ✓  | ✓     | -        | -   | -   | -   | -   |
| [9]  | ✓   | ✓  | ✓     | -        | -   | -   | ✓   | -   |
| [10] | ✓   | ✓  | ✓     | -        | -   | -   | -   | -   |
| [14] | ✓   | ✓  | ✓     | -        | -   | ✓   | -   | -   |
| [15] | ✓   | ✓  | -     | -        | -   | -   | ✓   | -   |
| [16] | ✓   | ✓  | ✓     | -        | -   | ✓   | -   | ✓   |
| [17] | ✓   | ✓  | ✓     | -        | ✓   | -   | ✓   | -   |
| [18] | -   | ✓  | ✓     | -        | ✓   | -   | -   | -   |
| [22] | ✓   | ✓  | ✓     | -        | ✓   | -   | ✓   | ✓   |
| [25] | ✓   | ✓  | ✓     | -        | -   | -   | ✓   | -   |
| Ours | ✓   | ✓  | ✓     | ✓        | ✓   | ✓   | ✓   | ✓   |

As mentioned above, extensive studies and different planning models have been directed to maximize the RES HC in the modern grid. However, most of these approaches focus on empowering either the HS or DR. It is worth mentioning that the simultaneous consideration of both HS and DR can provide further technical and economic benefits to the SGs, but requires precise modeling of a such complex problem with a low computational burden. A complete HS modeling is required that includes the hydrogen tank, water electrolyzer, and fuel cell to represent the real-time interactions with the grid. Other approaches adopt unrealistic assumptions for mitigating complexity, i.e., utilizing deterministic RES and load models, ignoring the reactive power support by RES inverters, and RES energy curtailment. In addition, many of the existing studies fail to incorporate the flexibility of EV charging stations in the planning model, yielding reduced benefits. In turn, these extra benefits must come with a more detailed modeling of the stochastic nature of EVs while empowering the controlled charging/discharging features. Most importantly, it is required to model the tradeoff between two competing objective functions: total costs (HS cost, RES cost, charging/discharging cost of EVs, DR discomfort cost, and curtailment costs of hydrogen and RESs) and HC of energy source in SGs.

To cover the research gap in the literature review, in this paper, the authors suggest a planning model to combine HS and DR in the best possible way to maximize the integration of intermittent photovoltaic and wind power in SGs. To accomplish this goal, a coordinated management scheme for multiple SG control devices is incorporated into the proposed planning model, which is missing in most existing approaches. To highlight the contribution of the paper, the major features of the proposed planning approach are compared with the previously planning approaches in Table I. The adopted control devices include 1) storage charging rates, 2) electrical DR (EDR), 3) hydrogen load DR (HDR), 4) RES energy curtailment options, and 5) DSTATCOM functionality of the RES interfacing inverters. Besides, a set of the control variables of a complete HS (including the HT, WE, and FC) are formulated in the planning model for SGs. Furthermore, precise modeling that allows for further RES and HS growth in SGs has been taken into consideration for the charging stations of EVs with G2V/V2G features. The suggested approach is

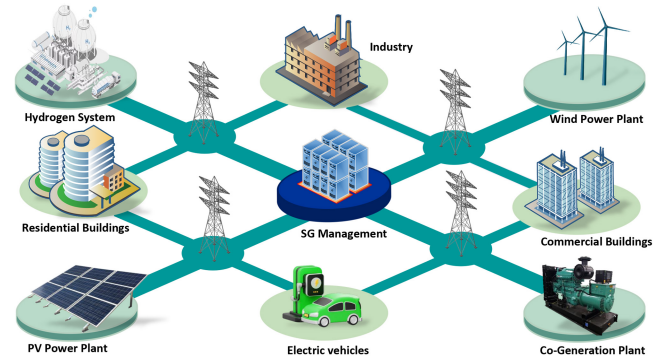


Fig. 1. A visual representation of a smart grid.

designed to maximize two conflicting objective functions—the overall costs and the HC of RESs using a multi-objective optimization framework. A mixed probability model that takes into account a realistic yearly dataset has been developed to reflect different uncertainties and give accurate locations and capacity for both hydrogen and renewable energy systems. Several simulation examples and comparisons with other models show the effectiveness of the suggested planning model. The results of the simulations demonstrate how well the suggested approach performs in providing high renewable energy penetration with appropriate HS at the lowest possible cost. An important extension for the proposed model has been highlighted, which is accurately defining the best number of HS in a certain SG to maximize RES size.

## II. PROBLEM DESCRIPTION

SGs are being formed based on an active network concept where various forms of energy sources and storage are integrated efficiently besides load centers. For instance, Fig. 1 depicts a visual representation of an SG in which PV and wind systems are considered energy sources supplying different load types (including flexible loads) while EV charging stations and HS are flexible energy storage systems. Such flexibility of distributed generation (Watt-VAr control functions), storage (charging/discharging flexibility), and load (DR options) can have significant positive impacts in improving SG performance with optimized operational costs. For this purpose, an SG management unit is considered which controls the energy flow and is responsible for energy generation, distribution, storage, and trading. In this work, we aim to empower such emerging SG flexibility to optimally plan RES units (PV and wind turbines) as well as HS in a simultaneous manner for efficient and economic operation.

## III. PROPOSED APPROACH

In this section, the mathematical formulation of the proposed multi-objective optimization model is presented. Maximizing the HC of RES is essential for a sustainable, low-carbon future, improving energy security, and driving economic growth. Additionally, it represents a pivotal step towards achieving climate goals, fostering cleaner environments, and advancing innovation in energy technologies. The proposed model aims to optimize the planning of RES and HS to



maximize the HC while minimizing the total annual cost. In this context, the HC and total annual cost are considered as conflicting objective functions in this study. It is worth mentioning that the primary goal is to develop an accurate and comprehensive model, even if it is computationally intensive. The various objective functions and constraints within this multi-objective optimization model can be expressed mathematically as follows.

#### A. Objective Functions

The proposed multiobjective optimization model aims to minimize two competing objective functions and can be mathematically formulated as follows:

$$\min_{\Theta} \{G_1, G_2\} \quad (1)$$

where  $\Theta$  represents the vector of the control variables and includes the locations and capacities of RES, locations and capacities of HS, hydrogen demand response, electrical demand response, reactive power of the RES interfacing inverters, and the G2V/V2G powers of the EVs.

The total annual cost objective function to be minimized by the proposed planning approach includes the investment and operation costs of RES and HS along with the cost of the electrical energy purchased/sold from/to the main electrical grid. Therefore, the cost objective function can be mathematically expressed by (2):

$$G_1 = \text{CRF} \times C^{\text{inv}} + \text{LF} \sum_{t \in \mathbb{T}} \sum_{\ell \in \mathbb{L}} \text{Nd}(t, \ell) \times C^{\text{ope}}(t, \ell) \times \Pi(t, \ell) \quad (2)$$

The CRF and LF can be described by (3) and (4), respectively.

$$\text{CRF} = \frac{\Re(\Re + 1)^{N_{\text{yr}}}}{(\Re + 1)^{N_{\text{yr}}} - 1} \quad (3)$$

$$\text{LF} = \frac{\mathcal{D}(\mathcal{D} + 1)^{N_{\text{yr}}}}{(\mathcal{D} + 1)^{N_{\text{yr}}} - 1} \times \frac{(1 + \mathcal{D}')^{N_{\text{yr}}} - 1}{\mathcal{D}'(1 + \mathcal{D}')^{N_{\text{yr}}}} \quad (4)$$

in which:

$$\mathcal{D}' = \frac{\mathcal{D} - \mathcal{E}}{1 + \mathcal{E}} \quad (5)$$

The investment cost can be mathematically formulated by (6).

$$C^{\text{inv}} = C^{\text{inv,HS}} + C^{\text{inv,PV}} + C^{\text{inv,WT}} \quad (6)$$

in which:

$$C^{\text{inv,HS}} = C^{\text{HS,HT}} + C^{\text{HS,FC}} + C^{\text{HS,WE}} \quad (7)$$

$$C^{\text{HS,HT}} = \left[ \sum_{i \in \mathbb{B}} N_i^{\text{HT}} \times \text{Pb}_i^{\text{HT}} \right] \times C^{\text{HT}} \quad (8)$$

$$C^{\text{HS,FC}} = \left[ \sum_{i \in \mathbb{B}} N_i^{\text{FC}} \times \text{Pb}_i^{\text{FC}} \right] \times C^{\text{FC}} \quad (9)$$

$$C^{\text{HS,WE}} = \left[ \sum_{i \in \mathbb{B}} N_i^{\text{WE}} \times \text{Pb}_i^{\text{WE}} \right] \times C^{\text{WE}} \quad (10)$$

$$C^{\text{inv,PV}} = \left[ \sum_{i \in \mathbb{B}} N_i^{\text{PV}} \times \text{Pb}_i^{\text{PV}} \right] \times C^{\text{PV}}, \quad (11)$$

$$C^{\text{inv,WT}} = \left[ \sum_{i \in \mathbb{B}} N_i^{\text{WT}} \times \text{Pb}_i^{\text{WT}} \right] \times C^{\text{WT}}, \quad (12)$$

On the other hand, the operation cost of the smart grid with RES and HS can be described by (13), shown at the bottom of the page. The second objective function to be minimized is the negative of the HC function, implying its maximization, and it can be formulated using (14).

$$G_2 = - \left( C^{\text{PV}} \times \sum_{i \in \mathbb{B}} N_i^{\text{PV}} + C^{\text{WT}} \times \sum_{i \in \mathbb{B}} N_i^{\text{WT}} \right) \quad (14)$$

#### B. Constraints

This subsection describes the various constraints associated with the planning problem, detailed as follows:

1) *Smart Grid Constraints:* The active and reactive power balance constraints can be described by (15) and (16) shown at the bottom of the next page, respectively.

In this work, the primary substation of the test system is assumed to be connected at bus  $i = 1$ . Hence, the active and reactive powers injected into the grid can be defined by (17)-(19).

$$P_i^g(t, \ell) = 0, \quad Q_i^g(t, \ell) = 0, \quad \forall i \neq 1, t \in \mathbb{T}, \ell \in \mathbb{L} \quad (17)$$

$$0 \leq P_i^g(t, \ell) \leq P_i^{g,\max}, \quad \forall i \in \mathbb{B}, t \in \mathbb{T}, \ell \in \mathbb{L} \quad (18)$$

$$0 \leq Q_i^g(t, \ell) \leq Q_i^{g,\max}, \quad \forall i \in \mathbb{B}, t \in \mathbb{T}, \ell \in \mathbb{L} \quad (19)$$

Furthermore, maintaining the voltage profiles at each bus and should adhere to designated minimum and maximum

$$\begin{aligned} C^{\text{ope}}(t, \ell) = & \underbrace{\pi^g(t, \ell) \times P^g(t, \ell)}_{\text{Grid operation cost}} + \underbrace{\pi^{\text{PV}} \sum_{i \in \mathbb{B}} P_i^{\text{PV}}(t, \ell)}_{\text{PV operation cost}} + \underbrace{\pi^{\text{WT}} \sum_{i \in \mathbb{B}} P_i^{\text{WT}}(t, \ell)}_{\text{WT operation cost}} + \underbrace{\pi^{\text{h2p}} \sum_{i \in \mathbb{B}} P_i^{\text{H2P}}(t, \ell) + \pi^{\text{p2h}} \sum_{i \in \mathbb{B}} P_i^{\text{P2H}}(t, \ell)}_{\text{HS operation cost}} \\ & + \underbrace{\pi^{\text{curt}} \left( \sum_{i \in \mathbb{B}} P_i^{\text{Wc}}(t, \ell) + \sum_{i \in \mathbb{B}} P_i^{\text{Pvc}}(t, \ell) \right)}_{\text{Curtainnet cost of WT and PV}} + \underbrace{\pi^{\text{ch}} \sum_{i \in \mathbb{B}} P_i^{\text{CH}}(t, \ell)}_{\text{Curtainnet cost of hydrogen}} + \underbrace{\pi^{\text{dr}} \sum_{i \in \mathbb{B}} |P_i^{\text{DR}}(t, \ell)|}_{\text{Discomfort cost of DR}} + \underbrace{\pi^h \sum_{i \in \mathbb{B}} P_i^{\text{H}}(t, \ell)}_{\text{Purchased hydrogen}} \\ & + \underbrace{\pi^{\text{hdr}} \sum_{i \in \mathbb{B}} |P_i^{\text{HDR}}(t, \ell)|}_{\text{Discomfort cost of HDR}} + \underbrace{\pi^{\text{EV}} \left( \sum_{i \in \mathbb{B}} \sum_{k \in \mathbb{N}} P_{i,k}^{\text{ch,EV}}(t, \ell) + \sum_{i \in \mathbb{B}} \sum_{k \in \mathbb{N}} P_{i,k}^{\text{dc,EV}}(t, \ell) \right)}_{\text{Operation cost of EVs}} - \underbrace{\pi^h \sum_{i \in \mathbb{B}} D_i^{\text{H}}(t, \ell)}_{\text{Hydrogen revenue}} \end{aligned} \quad (13)$$

limits outlined in the voltage regulations standards. This can be expressed mathematically as in (20)

$$V^{\min} \leq V_i(t, \ell) \leq V^{\max}, \quad \forall i \in \mathbb{B}, t \in \mathbb{T}, \ell \in \mathbb{L} \quad (20)$$

On the other hand, controlling the currents flowing through the lines is crucial to prevent thermal problems in the smart grid, as given in (21).

$$I_{ij} \leq I_{ij}^{\max} \quad (21)$$

2) *RES Constraints*: The combined capacities of PV and WT installations at each bus within the smart grid must remain within their designated limits, as specified by (22) and (23)

$$N_i^{\text{PV},\min} \leq N_i^{\text{PV}} \leq N_i^{\text{PV},\max}, \quad \forall i \in \mathbb{B} \quad (22)$$

$$N_i^{\text{WT},\min} \leq N_i^{\text{WT}} \leq N_i^{\text{WT},\max}, \quad \forall i \in \mathbb{B} \quad (23)$$

On the other hand, the reactive power capability of the RES interfacing inverters can be described (24)-(27)

$$\begin{cases} Q_i^{\text{PV},\max}(t, \ell) = \sqrt{(S_i^{\text{PV},r})^2 - (P_i^{\text{PV}}(t, \ell))^2} \\ Q_i^{\text{PV},\min}(t, \ell) = -\sqrt{(S_i^{\text{PV},r})^2 - (P_i^{\text{PV}}(t, \ell))^2} \end{cases} \quad (24)$$

$$\begin{cases} Q_i^{\text{WT},\max}(t, \ell) = \sqrt{(S_i^{\text{WT},r})^2 - (P_i^{\text{WT}}(t, \ell))^2} \\ Q_i^{\text{WT},\min}(t, \ell) = -\sqrt{(S_i^{\text{WT},r})^2 - (P_i^{\text{WT}}(t, \ell))^2} \end{cases} \quad (25)$$

Hence, it is imperative to ensure that the injected/absorbed reactive power from the RES inverters remains within these specified limits, as outlined in (26) and (27).

$$Q_i^{\text{PV},\min}(t, \ell) \leq Q_i^{\text{PV}}(t, \ell) \leq Q_i^{\text{PV},\max}(t, \ell) \quad (26)$$

$$Q_i^{\text{WT},\min}(t, \ell) \leq Q_i^{\text{WT}}(t, \ell) \leq Q_i^{\text{WT},\max}(t, \ell) \quad (27)$$

where equations (24)-(27) are  $\forall i \in \mathbb{B}, t \in \mathbb{T}, \ell \in \mathbb{L}$ .

3) *Constraints of HS*: The state of charge of the hydrogen tank (HSoC) at time instant  $t$  and state  $\ell$  can be formulated, as provided in (28).

$$\begin{aligned} \text{HSoC}_i(t, \ell) = & \text{HSoC}_i(t-1, \ell) + \Delta t \times (\eta^{\text{WE}} \times P_i^{\text{P2H}}(t, \ell) \\ & - \frac{P_i^{\text{H2P}}(t, \ell)}{\eta^{\text{FC}}} - D_i^{\text{H}}(t, \ell) + P_i^{\text{CH}}(t, \ell) \\ & - P_i^{\text{HDR}}(t, \ell)), \quad \forall i \in \mathbb{B}, t \in \mathbb{T}, \ell \in \mathbb{L} \end{aligned} \quad (28)$$

in which:

$$P_{i,\min}^{\text{P2H}} I_i^{\text{P2H}}(t, \ell) \leq P_i^{\text{P2H}}(t, \ell) \leq P_{i,\max}^{\text{P2H}} I_i^{\text{P2H}}(t, \ell), \quad (29)$$

$$P_{i,\min}^{\text{H2P}} I_i^{\text{H2P}}(t, \ell) \leq P_i^{\text{H2P}}(t, \ell) \leq P_{i,\max}^{\text{H2P}} I_i^{\text{H2P}}(t, \ell), \quad (30)$$

$$I_i^{\text{P2H}}(t, \ell) + I_i^{\text{H2P}}(t, \ell) \leq 1. \quad (31)$$

The HSoC should be kept within its minimum and maximum limits according to (32).

$$\text{HSoC}^{\min} \leq \text{HSoC}_i(t, \ell) \leq \text{HSoC}^{\max}, \quad \forall i \in \mathbb{B} \quad (32)$$

Equation (33) shows that the amount of hydrogen power stored in the HT at  $t = \{1, 24, 72, 96\}$  must be equal to the initial values.

$$\text{HSoC}_i(t, \ell) = \text{HSoC}_i^{\text{initial}}, \quad \forall t \in \{1, 24, 72, 96\} \quad (33)$$

The energy balance constraint for the HS is presented in (34),

$$\begin{aligned} P_i^{\text{P2H}}(t, \ell) = & D_i^{\text{H}}(t, \ell) + P_i^{\text{H2P}}(t, \ell) + P_i^{\text{HDR}}(t, \ell) \\ & - P_i^{\text{CH}}(t, \ell) - P_i^{\text{H}}(t, \ell) \end{aligned} \quad (34)$$

Moreover, the capacities of HT, FC, and WE are constrained as follows (35)-(37):

$$N_i^{\text{HT},\min} \leq N_i^{\text{HT}} \leq N_i^{\text{HT},\max}, \quad \forall i \in \mathbb{B} \quad (35)$$

$$N_i^{\text{FC},\min} \leq N_i^{\text{FC}} \leq N_i^{\text{FC},\max}, \quad \forall i \in \mathbb{B} \quad (36)$$

$$N_i^{\text{WE},\min} \leq N_i^{\text{WE}} \leq N_i^{\text{WE},\max}, \quad \forall i \in \mathbb{B} \quad (37)$$

4) *EV Constraints*: This subsection provides an explanation of the G2V and V2G models pertaining to EV batteries. The Changes in EV's State of Charge (SoC) during G2V and V2G processes at each time instant  $t$  and under state  $\ell$  can be outlined as follows (38):

$$\begin{aligned} \text{SoC}_{i,k}(t, \ell) = & \text{SoC}_{i,k}(t-1, \ell) + \eta^{\text{ch}} \times P_{i,k}^{\text{ch},\text{EV}}(t, \ell) \\ & \times I_{i,k}^{\text{ch},\text{EV}}(t, \ell) - \frac{P_{i,k}^{\text{dch},\text{EV}}(t, \ell) \times I_{i,k}^{\text{dch},\text{EV}}(t, \ell)}{\eta^{\text{dch}}}. \end{aligned} \quad (38)$$

To ensure that EV batteries do not charge and discharge simultaneously, a constraint is implemented to prevent this concurrent charging and discharging, as described in (39).

$$I_{i,k}^{\text{ch},\text{EV}}(t, \ell) + I_{i,k}^{\text{dch},\text{EV}}(t, \ell) \leq 1. \quad (39)$$

In efforts to prolong the lifespan of EVs and reduce degradation rates, a Depth of Discharge (DOD) of 80% is adopted in this work. Consequently, ensuring a minimum SOC of 20%. Thus, the daily driving distance should not surpass 80% of the maximum daily driving distance. This limitation can be expressed as follows (40):

$$20\% \leq \text{SoC}_{i,k}(t, \ell) \leq 100\%. \quad (40)$$

Conversely, the SoC of the EVs at the departure time ( $t_d$ ) must exceed or equal the minimum required value set by the owner. To enforce this requirement, the following constraint is implemented:

$$\text{SoC}_{i,k}(t_d, \ell) \geq \text{SoC}_{i,k}^{\min}. \quad (41)$$

$$\begin{aligned} P_i^{\text{S}}(t, \ell) + P_i^{\text{PV}}(t, \ell) + P_i^{\text{WT}}(t, \ell) + \sum_{k \in \mathbb{N}} P_{i,k}^{\text{dch},\text{EV}}(t, \ell) + P_i^{\text{H2P}}(t, \ell) - P_i^{\text{d}}(t, \ell) - \sum_{k \in \mathbb{N}} P_{i,k}^{\text{ch},\text{EV}}(t, \ell) - P_i^{\text{P2H}}(t, \ell) \\ - P_i^{\text{DR}}(t, \ell) - P_i^{\text{WC}}(t, \ell) - P_i^{\text{PVC}}(t, \ell) = V_i(t, \ell) \sum_{j \in \mathbb{B}} V_j(t, \ell) |Y_{ij}| \cos(\delta_{ij}(t, \ell) + \theta_{ji}(t, \ell)), \quad \forall i \in \mathbb{B}, t \in \mathbb{T}, \ell \in \mathbb{L} \end{aligned} \quad (15)$$

$$Q_i^{\text{S}}(t, \ell) + Q_i^{\text{PV}}(t, \ell) + Q_i^{\text{WT}}(t, \ell) - Q_i^{\text{d}}(t, \ell) = -V_i(t, \ell) \sum_{j \in \mathbb{B}} V_j(t, \ell) |Y_{ij}| \sin(\delta_{ij}(t, \ell) + \theta_{ji}(t, \ell)), \quad \forall i \in \mathbb{B}, t \in \mathbb{T}, \ell \in \mathbb{L} \quad (16)$$

5) *Electrical and Hydrogen Demand Response Constraints:* Demand Response (DR) programs are designed to reduce or shift electricity consumption during peak periods, thereby mitigating the need for constructing new assets exclusively utilized for limited hours annually. In this context, several DR programs can be employed for this purpose, such as price-based DR, and incentive-based DR [35]. In the first category, consumers are encouraged to adjust their consumption in response to price signals. This involves shifting their demand from peak to off-peak hours. Meanwhile, in the second category, consumers are motivated to engage in demand response programs through offered incentives. In this work, the first category is employed for electrical and hydrogen demands. The mathematical depiction of this DR program model is as follows:

$$P_i^{d,DR}(t, \ell) = P_i^d(t, \ell) + P_i^{DR}(t, \ell), \quad (42)$$

$$D_i^{H,DR}(t, \ell) = D_i^H(t, \ell) + P_i^{HDR}(t, \ell), \quad (43)$$

$$-P_i^d(t, \ell) \times P_{\max}^{DR} \leq P_i^{DR}(t, \ell) \leq P_i^d(t, \ell) \times P_{\max}^{DR}, \quad (44)$$

$$-D_i^H(t, \ell) \times P_{\max}^{HDR} \leq P_i^{HDR}(t, \ell) \leq D_i^H(t, \ell) \times P_{\max}^{HDR}, \quad (45)$$

in which:

$$\sum_{i \in \mathbb{B}} \sum_{t \in \mathbb{T}} P_i^{DR}(t, \ell) = 0, \quad (46)$$

$$\sum_{i \in \mathbb{B}} \sum_{t \in \mathbb{T}} P_i^{HDR}(t, \ell) = 0. \quad (47)$$

It is worth mentioning that, in this work,  $P_{\max}^{DR}$  and  $P_{\max}^{HDR}$  are considered to be 20% of the base load.

### C. Modeling Uncertainty in RES and Load Demand

This subsection models the uncertainties associated with RES load demand. We have considered a study period of one year, which is divided into four seasons, with a representative day generated for each season to capture the random behavior of various RES and load demand during each season. The Beta probability distribution function (PDF) is applied to represent solar irradiance variability, while the Rayleigh PDF characterizes the stochasticity of wind speed. Conversely, the normal PDF is used to capture the uncertainty in load demand [36], [37], [38]. The study leverages hourly historical data spanning three years encompassing solar irradiance, wind speed, and load demand. Each year is split into four seasons, with each season being represented by any day within that season. The data are then used to create a frequency distribution of the irradiance, wind speed, and load demand measurements for each season's typical day. Each day is divided into 24 hourly segments, resulting in a total of 96-time segments for the year (24 hours per season). Assuming each month has 30 days, each time segment contains 270 (3 months  $\times$  30 days/month  $\times$  3 years) solar irradiance, wind speed, and load demand data points. The mean and standard deviation for each time segment are calculated utilizing these data. Then, the Beta, Weibull, and normal PDFs are generated for each hour.

It is worth mentioning that the impact of changing environmental conditions and grid dynamics can introduce

uncertainties. However, considering such changing environmental conditions and grid dynamics, it can significantly increase the computational burden of the proposed planning approach. Accordingly, this assumption (historical data for probability modeling) is followed for the sake of applicability, where it is widely adopted in the literature [37], [39], [40].

The Beta PDF of the solar irradiance ( $\mathcal{R}$ ) at time instant ( $t$ ) can be described as follows:

$$f^b(t, \mathcal{R}) = \frac{\Gamma(\alpha(t) + \beta(t))}{\Gamma(\alpha(t)) \times \Gamma(\beta(t))} \mathcal{R}^{\alpha(t)-1} (1 - \mathcal{R})^{\beta(t)-1} \quad (48)$$

The parameters of the Beta PDF ( $\alpha$  and  $\beta$ ) can be calculated as follows:

$$\alpha = \frac{\mu_{\mathcal{R}}(t) \times \beta}{(1 - \mu_{\mathcal{R}}(t))} \quad (49)$$

$$\beta = (1 - \mu_{\mathcal{R}}(t)) \times \left( \frac{(1 - \mu_{\mathcal{R}}(t)) \times \mu_{\mathcal{R}}(t)}{(\sigma_{\mathcal{R}}(t))^2} \right) \quad (50)$$

The probability of solar irradiance for each state can be determined by (51), as follows:

$$\mathcal{P}^{\mathcal{R}}(t, y) = \int_{\mathcal{R}_{y1}}^{\mathcal{R}_{y2}} f^b(t, \mathcal{R}) \cdot d\mathcal{R} \quad (51)$$

After generating the Beta PDF for a specific time segment, the output power for the various states can be calculated for that segment, as discussed in [40].

The Weibull PDF of the wind speed ( $v$ ) at the time instant ( $t$ ) can be described as follows:

$$f^r(t, v) = k/c(v(t)/c)^{k-1} \exp[-(v(t)/c)^k] \quad (52)$$

The scale and shape parameters ( $c$ ) and ( $k$ ) can be calculated as follows [41]:

$$k = (\sigma_v(t)/\mu_v(t))^{-1.086} \quad (53)$$

$$c = \mu_v(t)/\Gamma(1 + (1/k)) \quad (54)$$

The occurrence probability of the wind speed for each state can be determined as follows:

$$\mathcal{P}^v(t, w) = \int_{v_{w1}}^{v_{w2}} f^r(t, v) \cdot dv \quad (55)$$

Then, the output power of the WT for the diverse states at a time segment can be computed [41].

On the other hand, the normal PDF of the load demand can be calculated by (56).

$$f^n(t, d) = \exp\left[-(l - \mu_d(t))^2 / 2(\sigma_d(t))^2\right] / (\sigma_d(t)\sqrt{2\pi}) \quad (56)$$

The probability of the load demand for the state  $l$  during time segment  $t$  is computed as follows:

$$\mathcal{P}^d(t, l) = \int_{d_{l1}}^{d_{l2}} f^n(t, d) \cdot dd \quad (57)$$

A comprehensive joint probability model is developed for solar irradiance, wind speed, and load demand to be utilized

in the proposed approach. This model incorporates the probabilities associated with solar irradiance, wind speed, and load demand, and can be expressed mathematically as in (58).

$$\Pi(t, \ell) = \mathcal{P}^{\mathcal{R}}(t, y) \times \mathcal{P}^v(t, w) \times \mathcal{P}^d(t, l) \quad (58)$$

Hence, the complete joint probability model can be established by accounting for all potential combinations of solar irradiance, wind speed, and load demand. Therefore, the entirety of the probability model can be outlined as follows:

$$\mathfrak{B} = [\{\mathcal{X}(t, \ell), \Pi(t, \ell)\}, \quad \forall t \in \mathbb{T}, \ell \in \mathbb{L}]. \quad (59)$$

#### IV. RESULTS AND DISCUSSIONS

This section describes the outcomes achieved by the proposed approach in comparison to existing approaches to demonstrate its superiority.

In this work, Multi-Objective Grey Wolf Optimization (MOGWO) combined with the General Algebraic Modeling System (GAMS) is employed to obtain the non-dominated Pareto optimal solutions, balancing the trade-offs between the objective functions (i.e.,  $\mathcal{G}_1$  and  $\mathcal{G}_2$ ).

Once MOGWO converges, the process identifies non-dominated Pareto optimal solutions. From these, a solution is selected that best aligns with the decision maker's preferences. However, the decision-maker's preferences are inherently uncertain. Therefore, it is assumed that the decision-maker has vague goals for each objective function. To address this, fuzzy set theory is employed as a framework [42]. The membership functions within fuzzy set theory denote the targets for each objective function. These functions are defined based on the decision maker's experiences and conjectural knowledge. The membership function can be characterized by:

$$\mu_{\zeta} = \begin{cases} 1, & f_{\zeta} \leq f_{\zeta}^{\min} \\ \frac{f_{\zeta}^{\max} - f_{\zeta}}{f_{\zeta}^{\max} - f_{\zeta}^{\min}}, & f_{\zeta}^{\min} < f_{\zeta} < f_{\zeta}^{\max} \\ 0, & f_{\zeta} \geq f_{\zeta}^{\max} \end{cases} \quad (60)$$

For any non-dominated solution  $m$ , the normalized membership function  $m$  can be computed as follows:

$$\mu^m = \sum_{\zeta=1}^{N_{obj}} \mu_{\zeta}^m / \sum_{m=1}^{N_{nd}} \sum_{\zeta=1}^{N_{obj}} \mu_{\zeta}^m \quad (61)$$

The solution that has maximum  $\mu^m$  membership in the fuzzy set is the best compromise solution.

##### A. Test System

This study employs the IEEE 69-bus distribution network displayed in Fig. 1 to assess the effectiveness of the proposed approach. Detailed line and bus data for the respected test system is given in [43]. In this work, three PV systems, three WT systems, and four HS are intended to be optimally allocated to maximize the HC and minimize the annual cost. Each unit of the PV and WT systems is considered to be 50 kW. Furthermore, the studied test system accommodates three charging stations (CSs) for EVs, which are connected to buses 13, 39, and 57. The data of the different CSs are

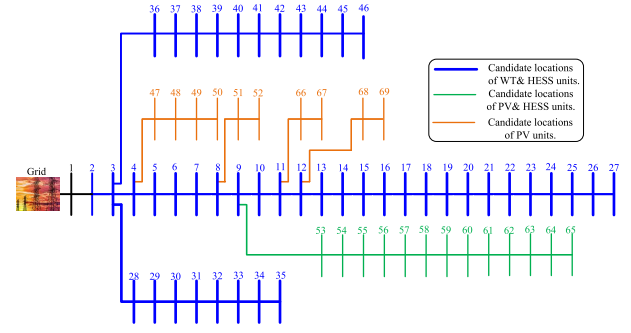


Fig. 2. The single line diagram of the IEEE 69-bus system.

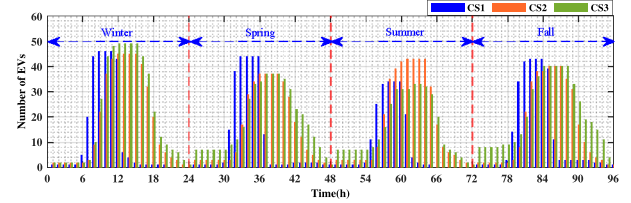


Fig. 3. The distribution of EVs of different CSs throughout the day varies across different seasons.

provided by the Toronto Parking Authority, Toronto, Ontario, Canada [44]. The influx of EVs to these CSs per hour throughout the day varies across different seasons following the distribution outlined in Fig. 3. Additionally, the initial State of Charge (SOC) for each EV conforms to the normal probability distribution function as described in [37]. The Tesla Model S batteries with a capacity of 85 kWh are employed in this work. Moreover, to model the uncertainties associated with RES and load, historical data spanning three years is utilized to generate the probabilities of solar irradiance, wind speed, and load demand. These probabilities are derived by representing each season of the three years as a day within these years. The simulations are executed within the MATLAB environment combined with GAMS software.

##### B. Assessing the Efficacy of the Proposed Approach

In this subsection, the performance of the proposed approach is evaluated. For this purpose, three different approaches are studied and compared with the proposed approach. These approaches are as follows:

- *Approach 1:* In this approach, the RES are optimally placed and sized to achieve the objective, considering the EVs along with their G2V and V2G technologies, as well as the active power curtailment (APC) capability of RES.
- *Approach 2:* Here, the PV and WT systems are strategically designed to maximize HC and minimize annual costs. This planning approach considers EVs with their G2V and V2G technologies, the APC capability of RES, along the DSTATCOM feature of the RES inverters.
- *Approach 3:* This approach is similar to Approach 2, while a DR program is employed for electrical loads.
- *Approach 4 (Proposed):* In this approach, the RES and HS, comprising HT, WE, and FC, are optimally allocated



TABLE II  
THE SIMULATION RESULTS OF THE DIFFERENT APPROACHES

| Approach                 | DG locations | DG size (MW) | Number of units | Max. Q (Mvar) | HT (MWh) | HC (MW) | Annual cost (M\$) | Cost reduction (%) | HC improvement (%) | Curtailed Energy (MWh) | DR (kW) |           |
|--------------------------|--------------|--------------|-----------------|---------------|----------|---------|-------------------|--------------------|--------------------|------------------------|---------|-----------|
|                          |              |              |                 |               |          |         |                   |                    |                    |                        | Max. up | Max. down |
| Approach 1               | PV           | 52           | 1.55            | 31            | -        | 8.7     | 2.11              | -                  | -                  | 140                    | -       | -         |
|                          |              | 56           | 1.75            | 35            | -        |         |                   |                    |                    |                        |         |           |
|                          |              | 66           | 1.05            | 21            | -        |         |                   |                    |                    |                        |         |           |
|                          | WT           | 10           | 1.75            | 35            | -        |         |                   |                    |                    |                        |         |           |
|                          |              | 29           | 0.85            | 17            | -        |         |                   |                    |                    |                        |         |           |
|                          |              | 37           | 1.75            | 35            | -        |         |                   |                    |                    |                        |         |           |
| Approach 2               | PV           | 47           | 2.00            | 40            | 0.561    | 9.7     | 1.74              | 17.5               | 11.50              | 124                    | -       | -         |
|                          |              | 58           | 2.00            | 40            | 1.293    |         |                   |                    |                    |                        |         |           |
|                          |              | 69           | 1.35            | 27            | 0.025    |         |                   |                    |                    |                        |         |           |
|                          | WT           | 9            | 1.75            | 35            | 0.981    |         |                   |                    |                    |                        |         |           |
|                          |              | 28           | 0.9             | 18            | 0.063    |         |                   |                    |                    |                        |         |           |
|                          |              | 37           | 1.75            | 35            | 0.119    |         |                   |                    |                    |                        |         |           |
| Approach 3               | PV           | 47           | 1.95            | 39            | 0.562    | 9.8     | 1.36              | 35.5               | 12.64              | 88.5                   | 339     | 397       |
|                          |              | 58           | 2.00            | 40            | 1.34     |         |                   |                    |                    |                        |         |           |
|                          |              | 68           | 1.65            | 33            | 0.046    |         |                   |                    |                    |                        |         |           |
|                          | WT           | 6            | 1.75            | 35            | 0.893    |         |                   |                    |                    |                        |         |           |
|                          |              | 28           | 0.75            | 15            | 0.063    |         |                   |                    |                    |                        |         |           |
|                          |              | 36           | 1.70            | 34            | 0.136    |         |                   |                    |                    |                        |         |           |
| Approach 4<br>(proposed) | PV           | 49           | 1.85            | 37            | 0.547    | 1.89    | 0.64              | 70.0               | 17.24              | 75.3                   | 254     | 249       |
|                          |              | 57           | 1.95            | 39            | 1.32     |         |                   |                    |                    |                        |         |           |
|                          |              | 66           | 1.55            | 31            | 0.025    |         |                   |                    |                    |                        |         |           |
|                          | WT           | 11           | 1.70            | 34            | 0.738    |         |                   |                    |                    |                        |         |           |
|                          |              | 31           | 1.45            | 29            | 0.032    |         |                   |                    |                    |                        |         |           |
|                          |              | 36           | 1.60            | 32            | 0.133    |         |                   |                    |                    |                        |         |           |

TABLE III  
DETAILED SIMULATION RESULTS OF THE PROPOSED APPROACH

|                           |               |      |      |      |     |
|---------------------------|---------------|------|------|------|-----|
| Location of HS            |               | 3    | 31   | 39   | 56  |
| Size of HT (kWh)          |               | 660  | 240  | 630  | 360 |
| Size of electrolyzer (kW) |               | 148  | 135  | 132  | 150 |
| Size of fuel cell (kW)    |               | 62   | 69   | 130  | 28  |
| Curtailed hydrogen (kWh)  |               | 263  | 241  | 250  | 263 |
| HDR                       | Max up (kW)   | 11.3 | 9.3  | 10.6 | 14  |
|                           | Max down (kW) | 18.5 | 12.9 | 18.9 | 18  |

to maximize the HC and reduce annual costs. This allocation takes into account EVs equipped with G2V and V2G technologies, the APC capability of RES, and the DSTATCOM feature integrated into the RES interfacing inverters. Additionally, this approach encloses curtailment and DR considerations related to the hydrogen demand.

The simulation results from the various approaches outlined in Tables II and III reveal notable trends. Comparing Approach 1 to Approaches 2 and 3, the HC is increased by 11.5% and 12.64%, respectively, alongside a reduction in annual costs by 17.50% and 35.50%. Conversely, significant enhancements in both HC (17.24% improvement) and annual cost reduction (70.0% decrease) are evident by applying the proposed approach. This emphasizes the considerable impact of implementing HS (i.e., 1.89 MWh) and demand response programs for electrical and hydrogen loads in maximizing HC while minimizing total annual costs. Furthermore, the total curtailed energy in the proposed approach (75.3 MWh) is lower than that in other approaches (140 MWh, 124 MWh, and 88.5 MWh). It can be noted from Table II that the locations, sizes, and maximum injected reactive power of the RES vary depending on the approach applied. Moreover, the maximum increase and decrease in load demand also vary with the approach used. For instance, with approaches 3 and 4, the maximum increase is 339 kW and

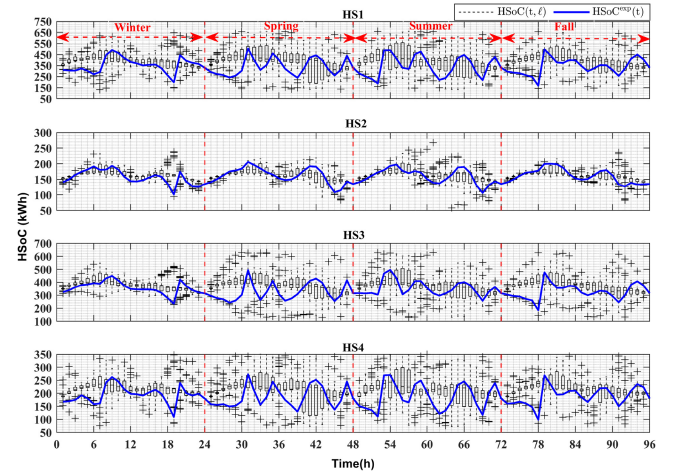
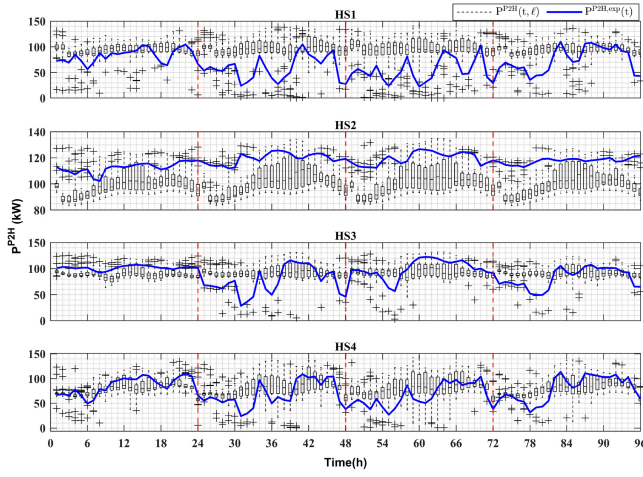
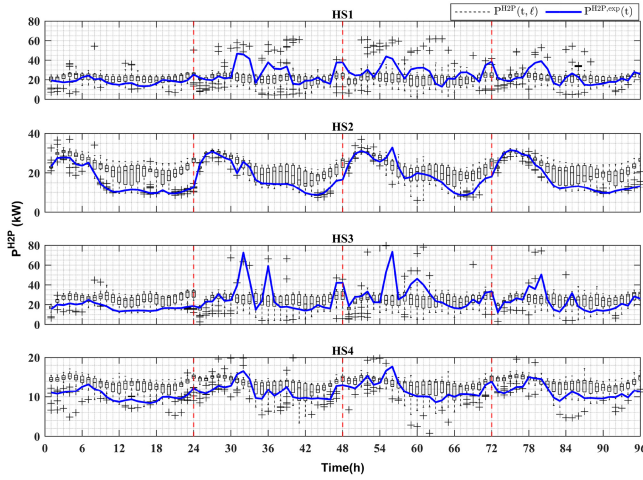
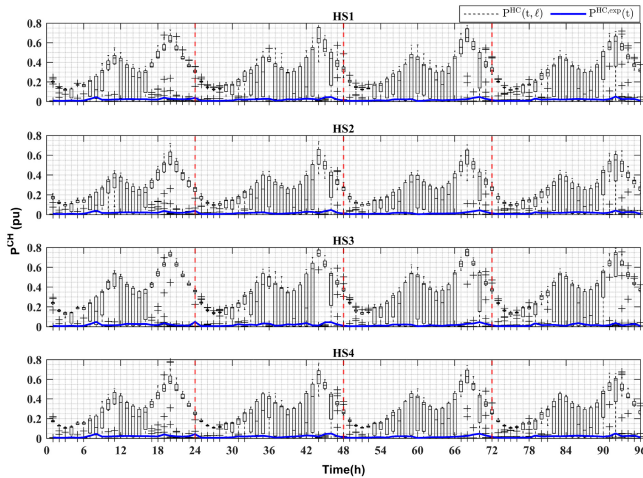


Fig. 4. State of charge of HTs at each time  $t$  and state  $\ell$  for different seasons.

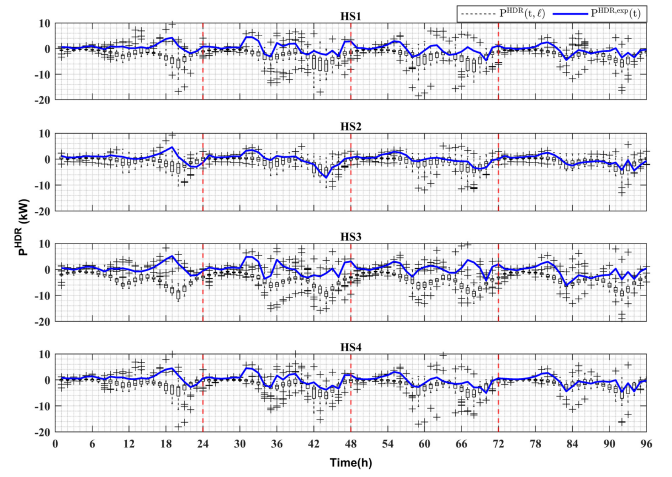
254 kW, respectively, while the maximum decrease is 397 kW and 249 kW, respectively. This indicates that the proposed approach reduces the need for significant load-shifting, thereby reducing discomfort costs.

Conversely, Table III displays the optimized locations and sizes of the HS components, including HTs, WEs, and FCs. The sizes of these assets vary in accordance with the HS location. Furthermore, in the proposed approach, the annual curtailed hydrogen amounts are 263 kWh, 241 kWh, 250 kWh, and 263 kWh at different locations. The maximum increases in hydrogen loads at different locations are 11.3 kW, 9.3 kW, 10.6 kW, and 14 kW, while the maximum decreases are 18.5 kW, 12.9 kW, 18.9 kW, and 18 kW.

Figs. 4–8 display the SOC for various HTs, power consumption by WEs for hydrogen generation (P2H), electrical power generation by FCs (H2P), transferred hydrogen loads via the DR program (up/down), and curtailed hydrogen loads. These figures present data for each time instant  $t$  and state  $\ell$ . The box

Fig. 5. The values of P2H at each time  $t$  and state  $\ell$  for different seasons.Fig. 6. The values of H2P at each time  $t$  and state  $\ell$  for different seasons.Fig. 7. The curtailed hydrogen loads at each time  $t$  and state  $\ell$  for different seasons.

plots illustrate values at each state, while solid lines indicate the expected values. From Figs. 4, 5, and 6, it is observable that the SOC of the HTs begins increasing from the start of the day in tandem with the rise in RES generation. This

Fig. 8. The transferred hydrogen loads down/up by the DR at each time  $t$  and state  $\ell$  for different seasons.

increase impacts both P2H and H2P: P2H escalates while H2P diminishes. However, after midday, the SOC of the HTs starts declining due to the waning RES generation. Consequently, P2H values decrease while H2P values rise, signifying the conversion of hydrogen back to electrical power to bolster the grid. This implies the pivotal role of the HS in enhancing the HC of the RES in the electrical networks. On the other hand, the curtailment of the hydrogen loads and RES along with the DR of the hydrogen and electrical demands can optimize hydrogen and RES utilization, as they ensure that the HS and RES are used more effectively. These optimal values of curtailment and DR (Figs. 7 and 10) contribute to better utilization of the resources and infrastructure. Enabling the DSTATCOM functionality in RES interfacing inverters offers several merits, including voltage regulation, reduction in line losses, and enhanced integration of RES into the grid. This, in turn, contributes to maximizing the HC of RES in distribution networks and reducing the total annual costs. Fig. 11 shows the expected values of the injected reactive power from different RES inverters.

On the other hand, optimal charging and discharging of EVs can play a pivotal role in maximizing HC and minimizing the annual costs in the context of electric grid management. By optimal charging of EVs, they can align with low-demand or high-renewable generation periods. Hence, the grid operators can avoid the need for costly infrastructure upgrades. Moreover, the integration of EVs as distributed energy storage systems supports RES incorporation. Fig. 12 shows the expected optimal charging and discharging powers of the EVs at different charging. It's noticeable that the discharging values are low because the parking periods in the used EV data are not sufficient for discharging to achieve the desired SoC at the departure time. Fig. 13 illustrates the non-dominated Pareto optimal solutions of the proposed approach, balancing the trade-offs between the HC and annual costs.

### C. Effect of HS Units on HC and Annual Cost

This subsection investigates the effect of the installed number of HS units in the smart grids on HC and annual costs,

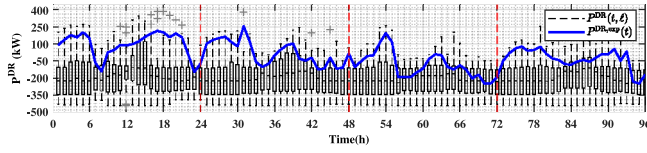


Fig. 9. The transferred electrical load down/up by the DR at each time  $t$  and state  $\ell$  for different seasons.

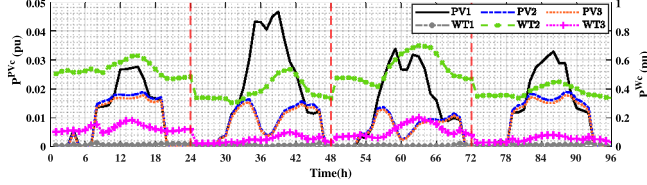


Fig. 10. The expected RES curtailment power at each time  $t$  for different seasons.

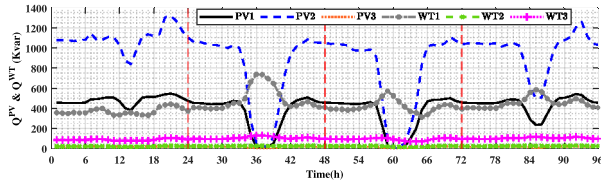


Fig. 11. The expected injected reactive power from RES inverters at each time  $t$  for different seasons.

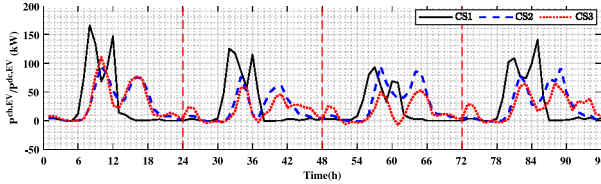


Fig. 12. The expected optimal charging/discharging power of the EVs at each time  $t$  for different seasons.

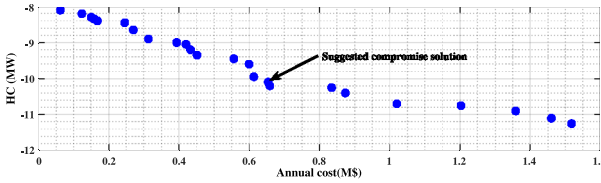


Fig. 13. The non-dominated Pareto optimal solutions of the proposed approach.

keeping the number of PV and WT systems the same (i.e., three PV and three WT systems). Fig. 14 illustrates how the changes in HC and annual costs correlate with the increasing number of installed HS units. This figure demonstrates that by increasing the installed number of HS, the annual costs significantly decrease, while the HC increases until reaching six HS units. Afterward, the annual costs start to increase again, and the HC decreases. The substantial reduction in annual costs can be attributed to hydrogen revenue. With each installed HS, a hydrogen load is connected. A portion of the stored hydrogen in the HS is converted back to electrical power to support the grid, while the remaining portion is sold to fulfill hydrogen loads. If more than six HS are installed, the hydrogen revenue will be lower compared to the investment

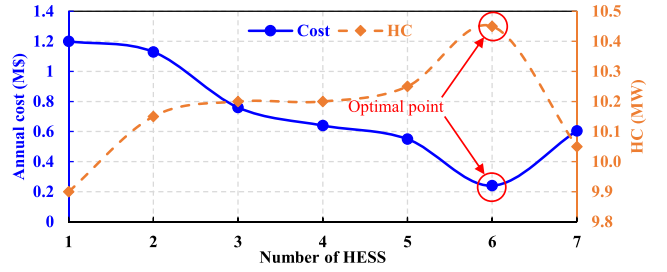


Fig. 14. The effect of the installed HS on HC and annual costs.

TABLE IV  
RESULTS OF THE IEEE 118-BUS SYSTEM USING THE  
PROPOSED APPROACH

|                          |       |       |       |       |       |       |
|--------------------------|-------|-------|-------|-------|-------|-------|
| PV Locations             | 18    | 56    | 87    | 115   |       |       |
| PV sizes (MW)            | 5.00  | 5.15  | 6.50  | 5.25  |       |       |
| Number of PV units       | 100   | 103   | 130   | 105   |       |       |
| Max. Q (Mvar)            | 0.64  | 1.02  | 1.34  | 0.47  |       |       |
| WT Locations             | 19    | 35    | 69    | 112   |       |       |
| WT sizes (MW)            | 5.00  | 4.85  | 7.35  | 4.75  |       |       |
| Number of WT units       | 100   | 97    | 147   | 95    |       |       |
| Max. Q (Mvar)            | 1.16  | 2.82  | 2.07  | 2.10  |       |       |
| HC (MW)                  | 43.85 |       |       |       |       |       |
| Annual cost (M\$)        | 2.66  |       |       |       |       |       |
| Curtailed energy (kWh)   | 65.62 |       |       |       |       |       |
| Location of HS           | 26    | 59    | 74    | 95    | 109   | 117   |
| Size of HT (MWh)         | 2.85  | 2.97  | 3.09  | 3.18  | 3.03  | 2.94  |
| Size of WE (kW)          | 296   | 299   | 290   | 287   | 289   | 266   |
| Size of FC (kW)          | 149   | 147   | 149   | 144   | 146   | 150   |
| Curtailed hydrogen (kWh) | 1.14  | 1.14  | 1.14  | 1.15  | 1.14  | 1.14  |
| HDR (Max. Up (kW))       | 26.81 | 26.78 | 26.83 | 26.87 | 26.86 | 26.83 |
| HDR (Max. Down (kW))     | 25.94 | 25.96 | 25.38 | 25.9  | 25.63 | 26.02 |
| DR (Max. up (MW))        | 4.34  |       |       |       |       |       |
| DR (Max. down (MW))      | 4.28  |       |       |       |       |       |

cost. This results in an increase in the total annual cost and, consequently, a decrease in the HC. These findings suggest that the optimal number of installed HS units in the system under study is six, achieving annual costs of 0.24 M\$ and HC of 10.45 MW.

#### D. Application of the Proposed Approach on a Larger Grid

In this subsection, the applicability of the proposed approach for maximizing RES HC while minimizing total annual costs in a large grid is evaluated. For this purpose, the proposed approach (Approach 4) is applied to the IEEE 118-bus systems [45]. This system consists of 118 loads and 117 branches, with total active and reactive loads of 22.71 MW and 17.04 Mvar, respectively. Here, four PV systems, four WT systems, and six HS are optimally allocated to maximize the HC and minimize the annual cost. Furthermore, the system under study includes six CSs for EVs, which are connected to buses {13, 39, 70, 92, 106, 115}.

Table IV presents the results of this case study. The table indicates that the total HC of the RES is 43.85 MW, while the total annual cost is 2.66 M\$. Additionally, the table details the optimal locations and sizes of various assets (i.e., PV, WT, HT, WE, and FC). The total curtailed energy from RES is 65.62 kWh, and the total curtailed hydrogen is 6.85 kWh. It is evident that both curtailed energy and curtailed hydrogen in this case study are lower compared to the previous case study. Furthermore, HT, WE, and FC capacities are higher. This capacity boosting implies that the system utilizes the RES



for hydrogen production and sells it to hydrogen customers or converts it back to electricity and injects it into the grid through the FC to gain more profit.

Accordingly, these results demonstrate that the proposed approach can be applied to larger grid networks, taking into account the complexities of managing diverse energy sources, demand response mechanisms, and grid infrastructures. This ensures that the proposed approach is both feasible and effective in real-world applications.

## V. CONCLUSION

This article proposed a planning model to optimally synergize HS and DR for enhanced integration of intermittent photovoltaic and wind generation in SGs interconnected with EV charging stations. Specifically, the proposed model is based on a coordinated management scheme for various SG control devices, encompassing the control variables of a complete HS, hydrogen load DR, RES energy curtailment options, DSTATCOM functionality of the RES interfacing inverters, storage charging rates, and electrical DR. The proposed model was formulated as a multi-objective optimization model that aimed to optimize two competing objective functions, namely total costs and hydrogen capacity of RESs. The efficacy of the proposed planning model was demonstrated through various simulation cases and comparisons with existing models. The simulation results revealed the superiority of the proposed model in optimally accommodating high RES penetration with proper HS at the lowest costs. For instance, the proposed model provided significant enhancements in both HC (17.24% improvement) with a 70.0% decrease in the annual cost compared to existing approaches. The proposed model was also applied to determine the optimal number of HS to maximize RES capacity in SGs. The proposed approach is a helpful tool for SG system planners, facilitating the optimal synergizing of HS and DR to enhance the integration of RES and EVs.

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